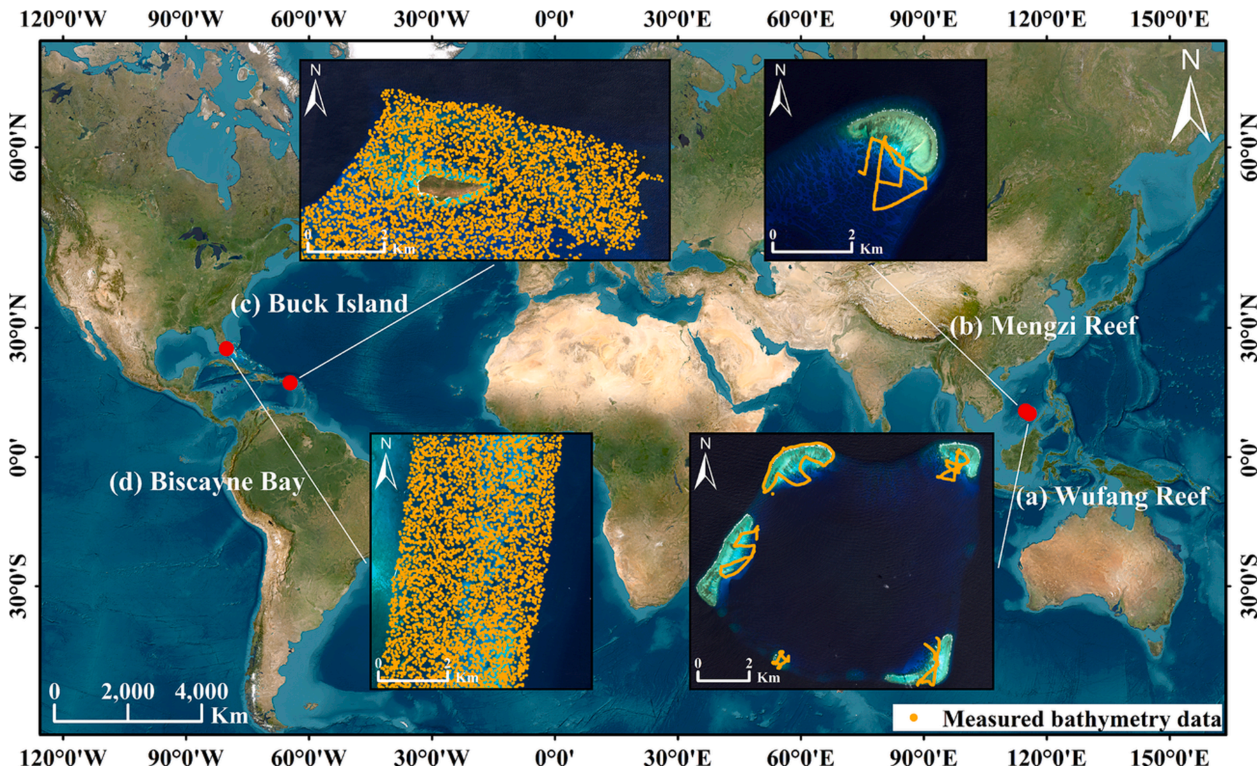

Fig. 1. Locations of the study areas and distributions of measured bathymetry data. (a) Wufang Reef, (b) Mengzi Reef, (c) Buck Island, (d) Biscayne Bay.

frequent cloud cover and rainfall. The locations of the study areas are shown in Fig. 1.

2.2. Satellite image data

The satellite image data used in this study is Sentinel-2 imagery obtained from the official website of the ESA. We downloaded and screened Sentinel-2 L1C images with less than 50 % cloud cover over the four study areas. In addition, to minimize the impact of changes in underwater topography, we used Sentinel-2 images acquired at similar times to the measured bathymetry data in the study areas, as detailed in Table 1.

2.3. Measured bathymetry data

The measured bathymetry data of Wufang Reef and Mengzi Reef was collected using a HD-MAX single-frequency sounder with a nominal accuracy of ± 1 cm + 0.1 % × depth. The coordinates of the survey points were obtained by Post Processed Kinematic (PPK) technology. The measured bathymetry data of Buck Island is a 1-m resolution Digital Elevation Model (DEM) generated from the Light Detection and Ranging (LiDAR) point data collected using a Riegl VQ-880-G II sensor, with a horizontal accuracy of 0.696 m at the 95 % confidence level and a vertical accuracy of 0.086 m. The measured bathymetry data of Biscayne Bay is also a 1-m resolution DEM that generated from the LiDAR point data collected using a Riegl VQ880G system, with a horizontal accuracy of 1.0 m at 1 sigma and a vertical accuracy of 0.15 m at 1 sigma. The measured bathymetry data for Buck Island and Biscayne Bay is available from NOAA Digital Coast (https://coast.noaa.gov/dataviewer/). The depth datum for all the measured bathymetry data is mean sea level (MSL). Details of the measured bathymetry data for the study areas are presented in Table 1.

3. Methodology

3.1. Satellite image data preprocessing

Atmospheric correction is a critical data preprocessing step in SDB. The ACOLITE model, developed by the Royal Belgian Institute of Natural Sciences (RBINS), is an atmospheric correction model designed specifically designed for aquatic environments and supports the correction of various medium- and high resolution remote sensing imagery. It includes two atmospheric correction algorithms, namely Exponential Extrapolation (EXP) (Vanhellemont and Ruddick, 2014; Vanhellemont and Ruddick, 2015) and Dark Spectrum Fitting (DSF) (Vanhellemont and Ruddick, 2018). The DSF exhibits robustness to variations in aerosol spatial distribution, making it more versatile and applicable in atmospheric correction scenarios. In recent years, ACOLITE has been widely used in SDB, with good applicability in both clear and turbid water environments (Caballero and Stumpf, 2020a; Caballero and Stumpf, 2021; Duan et al., 2022; Surisetty et al., 2021). Several recent studies have demonstrated that ACOLITE exhibits superior and more stable

Table 1
Specifics of the data used in this study.

Study Area	Location	Images Date	Image Number	Measured Bathymetry Date	Measured Bathymetry Sample
Wufang Reef	115.75° E, 10.50° N	2021.07–2022.12	33	2022.11	3285
Mengzi Reef	114.79° E, 11.15° N	2021.07–2022.12	36	2022.11	1010
Buck Island	64.61° W, 17.79° N	2019.01–2019.06	30	2019.01–2019.06	5000
Biscayne Bay	80.12° W, 25.51° N	2017.02–2018.01	26	2017.02	5000

performance in bathymetry compared to other atmospheric correction methods (Caballero and Stumpf, 2020a; Huang et al., 2023). In this study, the atmospheric correction of multi-temporal Sentinel-2 images is performed using the default DSF algorithm based on the officially provided ACOLITE software (https://odnature.naturalsciences.be/remsem/software-and-data/acolite).

3.2. Measured bathymetry data processing

Due to the difference in spatial resolution between sonar bathymetric points and Sentinel-2 imagery, multiple bathymetric points may correspond to a raster cell. The sonar bathymetric points of Wufang Reef and Mengzi Reef were rasterized to match the 10-m spatial resolution of the Sentinel-2 imagery. Specifically, an initial raster was created with the same spatial extent and resolution as the Sentinel-2 imagery in the visible bands, and the average depth of the bathymetric points falling within the raster cell was then used as the value for that cell. The measured bathymetry data for Buck Island and Biscayne Bay used in this study are 1-m resolution DEMs with different spatial resolutions and coordinate systems than the Sentinel-2 imagery. We conducted reprojection and resampling operations on the EDMs of the two study areas to obtain 10-m resolution DEMs that match the satellite imagery. The aforementioned processing of the measured bathymetry data is implemented based on the Geospatial Data Abstraction Library (GDAL). Subsequently, all measured bathymetric raster cells within 0–20 m depth range were used for bathymetric model calibration and accuracy assessment at Wufang Reef and Mengzi Reef, while 5,000 bathymetric raster cells within 0–20 m depth range were randomly selected and used at Buck Island and Biscayne Bay. The distribution of the measured bathymetry data used in this study is shown in Fig. 1. Finally, the measured bathymetry data for each study area was divided into control points and validation points according to the conventional ratio of 7:3.

3.3. Multi-temporal image weighted composition (MIWC) method

Due to the complexity and unpredictability of the remote sensing imaging process, remote sensing images are inevitably affected by various noise. Despite the effects of atmospheric absorption and scattering are eliminated by atmospheric correction, noise such as clouds and cloud shadows, bubble clouds, sun glint, and others can still significantly interfere with the bathymetric signal, as shown in Fig. 2 (a). Consequently, the noise may result in the generation of missing or unreliable bathymetric maps. Multi-temporal image composition is an effective way to solve the image noise problem. As illustrated in Fig. 2 (b), there is a discernible positive correlation between the reflectance of the visible bands and the NIR band of remote sensing images. This relationship has been demonstrated in previous studies and is widely applied to the removal of sun glint on water surfaces (Hedley et al., 2005; Hochberg et al., 2003). In addition, the reflectance of clouds and bubble clouds is significantly higher in the NIR band. Based on the principle, this paper proposes a NIR guided multi-temporal image weighted composition method. The main process of the method is shown in Fig. 3, with the following steps:

(1). Negative pixels mask. The negative pixels of the images in the atmospheric-corrected multi-temporal remote sensing image collection $C0$ are masked out to obtain the image collection $C1$.

(2). Iterative threshold segmentation of images. For the images in $C1$, an unsupervised water segmentation is performed on the NIR images using the Otsu algorithm. Specifically, The NIR reflectance R_{ij}^{NIR} is converted to 8-bit grayscale image to implement threshold segmentation based on OpenCV, then the threshold is inverse mapped to the numerical range of reflectance to get th_{ij}. The NIR reflectance of waters R_{ij}^{w-NIR} and its upper limit th_{ij}^{w} are obtained according to Eq. (1) and Eq. (2), respectively. The iterative threshold segmentation process is conducted for each image until th_{ij}^{w} is greater than the current waters segmentation threshold and less than the last segmentation threshold, and the current upper limit is added to the set of image segmentation thresholds T_{C1} (Eq. (3)).

$$R_{ij}^{w-NIR} = \left\{ R_{ij}^{NIR}(u,v) \middle| R_{ij}^{NIR}(u,v) \le th_{ij} \right\} \tag{1}$$

$$th_{ij}^{w} = QU\left(R_{ij}^{w-NIR}\right) + 1.5 \times \left(QU\left(R_{ij}^{w-NIR}\right) - QL\left(R_{ij}^{w-NIR}\right)\right) \tag{2}$$

$$T_{C1} = \left\{ th_{ij}^{w} \middle| th_{ij} \le th_{ij}^{w} \le th_{ij-1} \right\} \tag{3}$$

where i is the serial number of the image, j is the number of iterations for image segmentation,u and v are the pixel indices, QU and QL are the upper quartile and lower quartile, respectively.

(3). Calculation of the common segmentation threshold for waters. The upper and lower quartiles of T_{C1} are calculated, and the outlier thresholds and their corresponding images are removed according to Eq. (4) to obtain the segmentation threshold set T_{C2} and the corresponding image collection $C2$. Then the median value th_{C2}^{med} of the T_{C2} is taken as the common segmentation threshold for the images in $C2$. The aim is to weaken the effect of incomplete segmentation of waters that may occur in individual images and to ensure the similarity of reflectance in the subsequent image composition process as much as possible.

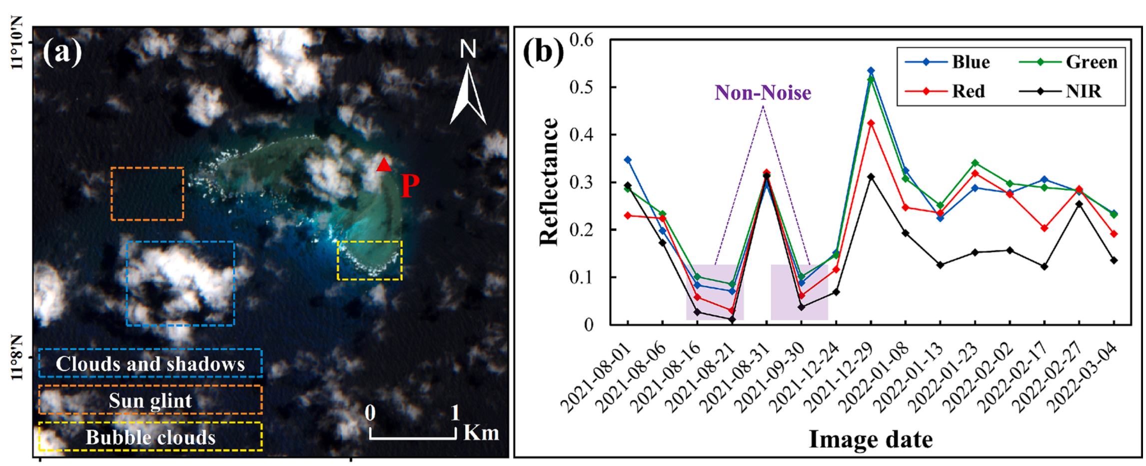

Fig. 2. Sentinel-2 image of Mengzi Reef. (a) Various noise on remote sensing image, (b) multi-temporal reflectance of the point P.

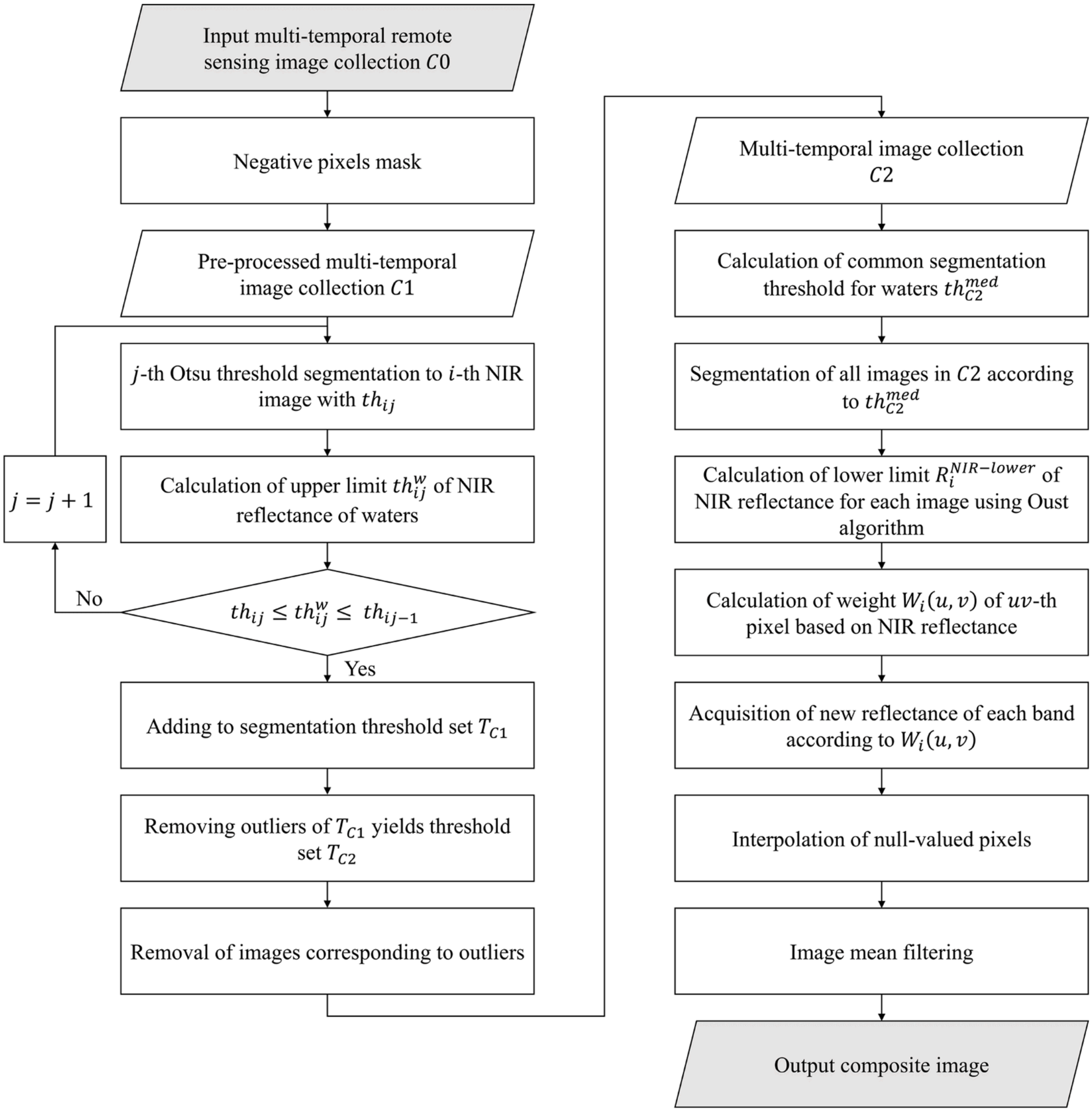

Fig. 3. Flow chart of the MIWC method.

$$\begin{cases} T_{C2} = \left\{ T_{C1}(i) \middle| q_{low} \leq T_{C1}(i) \leq q_{up} \right\}, i = 1, 2, 3, \ldots, n_{C1} \\ q_{low} = QU(T_{C1}) - 1.5 \times (QU(T_{C1}) - QL(T_{C1})) \\ q_{up} = QU(T_{C1}) + 1.5 \times (QU(T_{C1}) - QL(T_{C1})) \end{cases} \tag{4}$$

where q_{up} and q_{low} are the upper and lower bounds of the T_{C1}, and n_{C1} is the number of images in $C1$.

(4). Calculation of the lower limit of NIR reflectance. The images in $C2$ are segmented using the common threshold th_{C2}^{med} to obtain water body pixels for each image. Then, the NIR lower limit $R_i^{NIR-lower}$ for each image is determined by thresholding the NIR reflectance of the water body pixels using the Otsu algorithm.

(5). Multi-temporal image weighted composition. Multi-temporal image composition is performed pixel-by-pixel using the images in $C2$, and the weight $W_i(u,v)$ is calculated based on the NIR reflectance $R_i^{NIR}(u,v)$ of the $uv-$th pixel and the power parameter p(Eq. (5)). If the NIR reflectance is below the NIR lower limit of its corresponding image, the weight is calculated using the NIR lower limit. Finally, the new reflectance $R_{new}(u,v)$ of each band is obtained according to Eq. (6).

$$W_i(u,v) = \frac{1 \Big/ \left(\max\left(R_i^{NIR}(u,v), R_i^{NIR-lower}\right)\right)^p}{\sum_{i=1}^{n_{C2}} \left(1 \Big/ \left(\max\left(R_i^{NIR}(u,v), R_i^{NIR-lower}\right)\right)^p\right)}, \ if \ R_i^{NIR}(u,v) \leq th_{C2}^{med} \tag{5}$$

$$R_{new}(u,v) = \sum_{i=1}^{n_{C2}} (W_i(u,v) \times R_i(u,v)), \ if \ R_i^{NIR}(u,v) \leq th_{C2}^{med} \tag{6}$$

where n_{C2} is the number of images in $C2$. The recommended value of power parameter p is 4.

(6). Holes interpolation. For holes that may appear in the composite image, a linear interpolation is used to fill them.

(7). Mean filtering. To further suppress the possible residual noise and the neighboring pixel differences that appear after the image composition, a 3×3 mean filter is applied to the composite image.

3.4. Methods used for comparison

In order to evaluate the performance of the image composition method proposed in this paper, we conducted a comparative analysis between the MIWC method and the two existing image composition methods currently used in SDB, namely the median composition method (Li et al., 2021) and the maximum outlier removal method (Chu et al., 2019). In addition, the manually selected best image was included in the comparison. The parameter involved in the maximum outlier removal method was set to the recommended value. To ensure a fair comparison, a 3 × 3 mean filter was applied to each Sentinel-2 image.

3.5. Bathymetric model

We used two classical and widely used empirical bathymetric models to derive bathymetric maps and verify the effectiveness of the MIWC method proposed in this paper, including the log-transformed linear model (LLM) and the log-transformed ratio model (LRM). To eliminate the effect of potential residual noise on the image, a 3 × 3 median filter was applied to the SDB results.

(1). The LLM assumes that the ratio of bottom reflectance in the two bands in a given scene is consistent over all bottom types (Lyzenga, 1978; Lyzenga, 1985). In addition, the effects of atmospheric scattering and water surface reflections are eliminated by deep water correction. Its general form is given by Eq. (7).

$$Z = a_0 + \sum_{i=1}^{N} a_i \ln[R_w(\lambda_i) - R_\infty(\lambda_i)] \tag{7}$$

where Z is the water depth, N is the number of spectral bands used in the model, R_w is the reflectance of the water, $R_\infty(\lambda_i)$ is the reflectance of the deep water, a_0 and a_i are the model regression parameters to be solved. According to the previous studies (Shen et al., 2019; Su et al., 2015), green and blue bands with strong penetration capabilities are used for the LLM in this study.

(2). The LRM is based on the principle that the radiation attenuation rate differs in different bands, and uses the ratio of reflectance between two bands to derive the water depth with the following expression (Stumpf et al., 2003).

$$Z = m_1 \frac{\ln(nR_w(\lambda_i))}{\ln(nR_w(\lambda_j))} - m_0 \tag{8}$$

where n is a fixed constant value to ensure the logarithms are positive. m_0 and m_1 are the model regression coefficients to be solved. The LRM usually uses the ratio of blue band and green band for bathymetry (Caballero and Stumpf, 2020a; Stumpf et al., 2003), with n taken as 1000 (Stumpf et al., 2003), and this study follows this setup.

3.6. Accuracy evaluation metrics

In this study, the accuracy of SDB is evaluated using three metrics: the coefficient of determination (R^2), Bias, and the root mean square error (RMSE). These metrics are calculated as Eq. (9), Eq. (10) and Eq. (11), respectively.

$$R^2 = \frac{\left(\sum_{i=1}^{n}(x_i - \overline{x}_i)(y_i - \overline{y}_i)\right)^2}{\sum_{i=1}^{n}(x_i - \overline{x}_i)^2 \sum_{i=1}^{n}(y_i - \overline{y}_i)^2} \tag{9}$$

$$Bias = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_i) \tag{10}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2} \tag{11}$$

where x_i and y_i represent the measured and estimated bathymetry values, respectively, and n represents the number of validation points.

4. Results

4.1. Bathymetry results based on the images composited by the MIWC method

We used the MIWC method proposed in this paper to obtain composite images covering the four study areas based on the multi-temporal Sentinel-2 imagery, and the results are shown in Fig. 4. It can be seen that the composite images of the four regions are overall natural and free of significant noise, and the bottom types in the shallow waters are clearly visible. It is worth noting that the MIWC method effectively removed the bubble clouds that often appear at the edge of the reef plates, resulting in a complete display of the reef edge on the composite images. The results demonstrate that the MIWC method can effectively produce the high-quality optical images based on the Sentinel-2 multi-temporal images.

Bathymetric inversion was performed using the two bathymetric models, namely LLM and LRM, and the accuracy of the derived bathymetric maps from the composite image generated by the MIWC method was evaluated. As shown in Figs. 5 and 6, all the bathymetric maps derived from the composite images exhibit no significant outliers. The RMSE values of bathymetry using the LLM are 1.22 m, 0.73 m, 0.77 m and 0.67 m, and the RMSE values of bathymetry using the LRM are 1.23 m, 0.76 m, 0.84 m and 0.71 m in Wufang Reef, Mengzi Reef, Buck Island and Biscayne Bay, respectively. With the exception of Wufang Reef, high accuracies are achieved in all other study areas, with RMSE values below 1 m. As illustrated in Fig. 7, a small number of outliers are present in the bathymetry results of Wufang Reef that are significantly shallower than the measured bathymetry, resulting in the large bathymetry RMSE for Wufang Reef. Nevertheless, these results are still acceptable. In the four study areas, the R^2 values of all the bathymetry results are greater than or equal to 0.93. Meanwhile, it can also be seen from the scatterplots that the scatter points are basically closely distributed on both sides of the $y = x$ reference line, with the regression slopes are close to 1 and the regression intercepts are close to 0, indicating that the estimated depths are in good agreement with the measured depths. In addition, the absolute values of Bias of the bathymetry results for each study area are within 0.15 m, which indicates that there is no obvious systematic bias in these results. It demonstrates that satisfactory bathymetry results can be obtained from the composite images generated by the MIWC method, thereby proving the effectiveness of the image composition method proposed in this paper.

4.2. Comparison of the MIWC method with other methods

Figs. 8 (a) and 9 (a) show the best single Sentinel-2 images of Wufang Reef and Buck Island, acquired on November 4, 2022 and January 5, 2019, respectively. It is obvious that some clouds and cloud shadows are still present in the best single images, especially in the shallow waters situated to the southeast of Wufang Reef (Fig. 8 (g)) and the southwest of Buck Island (Fig. 9 (g)). In addition, significant bubble clouds are present on the best single image of Wufang Reef (Fig. 8 (a) and (g)), which are mainly found at the edges of the reef plate. Unlike clouds and cloud shadows, which are randomly distributed, bubble clouds are generally found at the edges of very shallow waters. The reflectance of these two types of noise is so high that it completely masks the normal optical signals of the water column. Additionally, the best single image of Wufang Reef exhibits a degree of sun glint, which affects the visualization of the seafloor. From Figs. 8 and 9, the median composition method and the maximum outlier removal method eliminate clouds, cloud shadows, and bubble clouds from the original images to some extent, but there are still obvious traces of clouds present in the composite images. In Buck Island, the median composition method and the maximum outlier removal method performed poorly, with a large number of residual noise traces showing patchy and dense distributions, resulting in unnatural composite images, especially for the maximum outlier